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Strategy-dependent coding of reward and delay during temporal discounting in macaque dorsolateral prefrontal cortex

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Corresponding Author:	Rossella Falcone Albert Einstein College of Medicine Bronx, UNITED STATES OF AMERICA
First Author:	Rossella Falcone
Order of Authors:	Rossella Falcone
	Siyu Wang
	David B. Weintraub
	Mariko McDougall
	Mark A. G. Eldridge
	Barry J. Richmond
Abstract:	Some rewards are worth the wait, while others are not. This behavior is governed by temporal discounting, whereby delays reduce the value of future rewards, such that only sufficiently large rewards justify long waits. Individuals differ widely in their strategies to weigh temporally delayed rewards. Prefrontal neurons are known to encode reward and delay; however, it remains unclear whether these neural representations reflect a generic temporal discounting computation process, or whether they depend on the individual behavioral strategies for trading off reward and time. We trained two monkeys to choose whether to accept or refuse delayed reward offers by releasing a lever at different times contingent on a go signal. We analyzed how the unforeseen go signals, which introduced different additional delays before an offer can be accepted, changed animals' temporal discounting decisions. Behavioral modeling revealed that the two monkeys adopted distinct strategies in re-evaluating the original reward offer in response to the go signal. Monkey S re-calculated the discounted value of the original offer by accounting for the additional delay associated with the go signal, whereas Monkey T treated the go signal as having additive values independent of the original offer. Although neurons in the dorsolateral prefrontal cortex (dlPFC) of both monkeys encoded reward and delay, we found important differences in how these neurons represented these factors, which reflected monkeys' respective behavioral strategies. Our results revealed a temporal discounting strategy-dependent coding of reward and delay in dlPFC.

Albert Einstein College of Medicine
Dr. Rossella Falcone, PhD
Assistant Professor
1300 Morris Park Ave
Kennedy Center, 429, Bronx, NY 10461
rossella.falcone@einsteinmed.edu

Dear Editor,

We are submitting our manuscript entitled *"Strategy-dependent coding of reward and delay during temporal discounting in macaque dorsolateral prefrontal cortex"* for consideration in *Progress Neurobiology*..

Temporal discounting, that delays reduce the value of future rewards, is often studied as a static preference for future rewards with known delays. In real-life however, either the expected delay or reward might oftentimes change amid the delay, requiring re-evaluation. (e.g., an unforeseen car crash on a highway can drastically change the time estimate for reaching a reward destination). In this study, we investigated how animals on-the-fly adjust their temporal discounting decisions when the originally expected delays change.

From both a model-free and model-based analysis, we showed that our animals adopted distinct strategies in response to changes in the temporal discounting offer. Neural analysis further revealed important differences between animals that agreed with their respective behavioral strategies. Although it is well established that dorsolateral prefrontal cortex (dlPFC) encodes reward and time during temporal discounting, our paper showed that instead of reflecting a generic temporal discounting computation process shared across individuals, dlPFC neurons represent reward and time in a way that is dependent on the individual strategies used for making temporal discounting decisions.

Given the broad interest of temporal discounting behavior, and the increasing interest for characterizing neural underpinnings of individual differences, we believe that our paper would be of interest to many and would fit the scope of *Progress Neurobiology*.

Best Regards,

Dr. Rossella Falcone, PhD

Abstract

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Title: Strategy-dependent coding of reward and delay during temporal discounting in macaque dorsolateral prefrontal cortex

Authors: Rossella Falcone[1,3,†,*], Siyu Wang[1,†], David B. Weintraub[2], Mariko McDougall[1], Mark A. G. Eldridge[1,4], and Barry J. Richmond[1,5].

Author affiliations:

[1]Laboratory of Neuropsychology, NIMH, Bethesda, MD, USA

[2]Surgical Neurology Branch, NINDS, Bethesda, MD, USA

[3]Leo M. Davidoff Department of Neurological Surgery, Albert Einstein College of Medicine Montefiore Medical Center Bronx, NY USA.

[4]Newcastle University, Newcastle upon Tyne, NE1 7RU.

[5]Senior Investigator, Emeritus.

†Equal contribution.

***Corresponding author:**

Rossella Falcone, PhD

Assistant Professor

Albert Einstein College of Medicine, Bronx, NY, 10461

e-mail: rossella.falcone@einsteinmed.edu

Abstract

Some rewards are worth the wait, while others are not. This behavior is governed by temporal discounting, whereby delays reduce the value of future rewards, such that only sufficiently large rewards justify long waits. Individuals differ widely in their strategies to weigh temporally delayed rewards. Prefrontal neurons are known to encode reward and delay; however, it remains unclear whether these neural representations reflect a generic temporal discounting computation process, or whether they depend on the individual behavioral strategies for trading off reward and time.

We trained two monkeys to choose whether to accept or refuse delayed reward offers by releasing a lever at different times contingent on a go signal. We analyzed how the unforeseen go signals, which introduced different additional delays before an offer can be accepted, changed animals' temporal discounting decisions. Behavioral modeling revealed that the two monkeys adopted distinct strategies in re-evaluating the original reward offer in response to the go signal. Monkey S re-calculated the discounted value of the original offer by accounting for the additional delay associated with the go signal, whereas Monkey T treated the go signal as having additive values independent of the original offer. Although neurons in the dorsolateral prefrontal cortex (dlPFC) of both monkeys encoded reward and delay, we found important differences in how these neurons represented these factors, which reflected monkeys' respective behavioral strategies. Our results revealed a temporal discounting strategy-dependent coding of reward and delay in dlPFC.

Introduction

Temporal discounting, the tendency to assign less value to delayed rewards (1, 2), is central to many everyday decisions, such as deciding whether to wait in line to dine at a restaurant. People differ widely in their temporal discounting preferences. For example, the “marshmallow” test (3) showed that some children can resist immediate gratification, while others succumb to it. Similar heterogeneity has been observed in animals (4-6). This diversity in temporal discounting preferences reflects the diverse strategies individuals adopt for trading off reward and time.

Temporal discounting preferences are usually considered static traits of an individual. Individual differences in temporal discounting behavior have been associated with a wide range of cognitive factors (7-9), changes in environmental conditions (10-12), and abnormal behaviors (13-16). How individuals adjust their temporal discounting behavior when the expectations for the delay or the reward changes (e.g., a person’s decision to wait in line for a restaurant table may change if a large group of people in front of them suddenly decided to leave the line) is less well studied.

The dorsal lateral prefrontal cortex (dlPFC) encodes variables important for making temporal discounting calculations, such as reward size and delay. In animals studies, it has been suggested that dlPFC explicitly supports temporally discounted value computations via a shared value coding dimension across magnitude and delay (17, 18). In humans, the involvement of dlPFC during intertemporal choice is also well documented (19, 20). However, it remains unclear whether the dlPFC coding of reward and time reflects a universal temporal discounting process shared across individuals, or whether it depends on the specific strategies used to compute discounted values.

In this work, we trained two monkeys to perform a two-step temporal discounting task, in which they chose between accepting or refusing a delayed reward offer. Crucially, to indicate an accept vs reject choice, monkeys had to release a lever in one of two different time windows, contingent on a go signal. In theory, the additional delay associated with different go signals modifies the discounted value of the original delayed reward offer. This allowed us to study how animals re-evaluated a delayed reward offer when an additional delay was introduced by the go signal. Behavioral modeling showed that our monkeys adopted different strategies. Neural analysis showed that, although dlPFC neurons of both animals encoded reward size and delay, their neural coding reflected the distinct behavioral strategies adopted by each animal.

Results

Two monkeys performed a two-step temporal discounting task (see Fig. 1A and Methods), in which they chose whether to accept or reject an offer of a delayed reward. Animals were pretrained on nine visual cues (Fig. 1B) that signaled different combinations of reward size (2, 4, or 6 drops of liquid reward) and delay (1, 5, or 10 seconds). Animals indicated their choice - accept or reject - by releasing a lever when a purple or yellow go signal, respectively, appeared. In each trial, after the reward offer was presented, one of the yellow/purple go signals (randomly drawn) was presented first. If monkey did not respond to the first go signal, the other go signal was presented after a fixed delay (1100ms). For instance, if the purple go signal appeared first, the animal could either release the lever immediately to accept the offer, or hold the lever and wait for the yellow go signal and then release the lever to refuse the offer. Conversely, if the yellow go signal appeared first, the animal could either release the lever immediately to refuse the offer, or wait for the purple go signal to accept the offer (Fig. 1A). If the animal accepted the offer, the cue-defined drops were delivered after the corresponding delay. If an offer was rejected, the next trial began immediately with a new offer chosen randomly among the nine available.

1. The go signal alters the temporal discounting behavior.

Across all the neural recording days (7 for Monkey T, 16 for Monkey S), both monkeys showed similar patterns of choice consistent with temporal discounting. They were more likely to accept offers of larger rewards with short delays, and less likely to accept smaller rewards with long delays (Fig. 2A-B). This temporal discounting behavior was modelled using a hyperbolic discounting function to compute the discounted value, and a logistic function to compute the acceptance probability as a function of the discounted value (Fig. 2A-B; see Methods).

In our task, to accept an offer, monkeys must release the lever when the purple go signal appears. If the first go signal is purple, animals can release the lever and accept the offer immediately. However, if the first go signal is yellow, animals have to wait an additional delay (1100ms) until the purple go signal appears to accept the offer. This additional delay, when the yellow go signal appears first, effectively increases the waiting time and reduces the offer value. Consequently, we expected that animals would have a lower probability of accepting the offer when yellow go signal shows first. Indeed, both animals showed lower probability of accepting an offer when the first go cue was yellow compared to purple (Fig. 3A-B).

2. Animals show distinct strategies for adjusting temporal discounting behavior in response to the go signal.

Although Monkey S rarely accepted the worst offers (e.g., 2 drops after 10 seconds) regardless of whether the first go signal was yellow or purple (Fig. 3A), Monkey T only showed low probabilities of accepting the worst offers when the first go signal was yellow (Fig. 3B). Interestingly, Monkey T accepted all offers at rates above 80% when the first go signal was purple (Fig. 3B).

The behavioral differences between monkeys prompted us to investigate the different strategies that can be used to adjust temporal discounting behavior based on the go signal. The baseline strategy (Model 1; see Methods) ignores the go signal and maintains a fixed acceptance preference for each of the nine offers. Clearly, neither of our monkeys used this strategy, but we included it as a control. The second simplistic strategy (Model 2; see Methods) assigns values directly to the go signal, and chooses to accept if the sum of the delayed reward offer value and the go signal value exceeds a threshold, and chooses to reject otherwise. The go signal value is an approximation of how much the go signal increases or decreases the offer value, and assumes that this increase or decrease is independent of the offer. In other words, the values of good and bad offers are increased or decreased by the same amount associated with the go signal. The third, optimal strategy (Model 3; see Methods), accounts for the delay associated with the go signal when re-calculating the temporally discounted value. The yellow and purple go signals modify the offer value based on the additional delay introduced by each go signal.

We fitted all three models to the monkey behavior. As expected, model comparison showed that Model 1, which assumed independent preference for each offer regardless of the go signal, was the worst fit for both animals. Model 3 best explained Monkey S's behavior, suggesting a recalculation of the discounted value when accounting for the go signal induced delay (Fig. 4A and 4C). In contrast, Monkey T's behavior was best explained by Model 2 (Fig. 4B and D), suggesting the use of a simpler heuristic. These models suggest that the two animals employed distinct strategies to adjust temporal discounting behavior based on the go signal.

3. The inverse coding of reward and delay in dlPFC reflects the temporal discounting computation.

First, we examined single neuron responses to reward and delay. We recorded 170 neurons in dlPFC (107 neurons in Monkey T and 63 neurons in Monkey S; Fig. S1A) while the animals performed the task. During the offer period, 121 out of 170 neurons were significantly modulated by reward magnitude and/or delay (2-way ANOVA, $p < 0.05$; Fig. S1B). Sliding-window ANOVA analysis showed that the proportion of neurons modulated by each factor increased shortly after cue presentation (Fig. S1C-D). To quantify the strength of encoding for each variable (reward magnitude, delay, and their interaction), we computed omega squared (ω^2), which estimates the proportion of variance in neural activity explained by each factor (see Methods). Neither monkey showed a significant representation of the upcoming choice (accept vs. refuse) following the offer onset (Fig. S1C-D and Fig. S2A-B, yellow lines), suggesting that dlPFC activity during this epoch – during which choice order was still unknown - primarily reflected the offer evaluation rather than the commitment to a choice.

Next, we defined the population coding dimensions of reward and delay based on the beta coefficients obtained from the sliding-window ANOVA analysis. For both animals, prefrontal neurons showed an inverse correlation between the encoding of reward and delay, reflecting temporal discounting computation (Fig. 5A-B). This relationship became stronger as the offer epoch progressed (Fig. S3). As a negative control, we performed the same analysis in the 250 ms window before cue onset , and no significance was found (Fig. S3A, E), suggesting that these relationships were unique to the evaluation period.

Although the inverse coding was present in both monkeys, we observed differences in the timing and duration of the inverse relationship between the two monkeys. In Monkey S, the inverse relationship emerged rapidly within 250ms after cue onset, and persisted throughout cue presentation (Fig. S3B–D). In Monkey T, a similar trend was observed, but it did not reach statistical significance during the early epochs (Fig. S3G–I).

4. Neural signatures of the go signal modulation of value encoding.

Behaviorally, we observed that monkeys' acceptance rate for offers was modulated by the go signal. Specifically, the animals were more likely to accept an offer if the first go signal was purple. We then looked for neural signatures of the go signal.

If the temporally discounted value is adjusted based on the additional delay introduced by the go signal (see behavioral Model 3 in the Methods), then compared to the purple go signal, which allows immediate acceptance of the offer, the yellow go signal, which requires an additional delay, should decrease the discounted value of the original offer. Therefore, we predicted that the population encoding dimension for the yellow versus purple go signal should be negatively aligned with the discounted value coding dimension (i.e., positively aligned with the reward size coding dimension, and negatively aligned with the delay coding dimension). We observed this pattern in Monkey S (Fig. 6A and 6C), but not in Monkey T (Fig. 6B and 6D).

This difference in neural encoding reflected the distinct behavioral strategies employed by the two monkeys, that only Monkey S adjusted the temporally discounted value based on the go signal; Monkey T did not.

5. Differences in the linearity of coding between monkeys

Differential behavioral and neural responses to the go signal reflect the different neural computations adopted by the two monkeys. We examined neural differences in value coding between monkeys at both the single-neuron and population levels.

At the single-neuron level, we tested whether firing rates during the first 750ms of the offer period were linearly modulated by the discounted value inferred from the behavioral model 1. Only 6 out of 63 neurons (10%) in Monkey S showed a significant linear correlation between spike count and discounted value (Fig. S4B), whereas in Monkey T, 35 out of 107 neurons (33%) exhibited linear coding (Fig. S4A). These results suggest that the dlPFC neurons in Monkey S exhibited more heterogeneous or non-linear patterns of value representation, whereas neurons in Monkey T encoded the value of the offer more linearly.

At the population level, principal component analysis revealed qualitative differences in condition-averaged neural trajectories between animals. Consistent with the single-neuron analysis, in Monkey S, the trajectories were entangled reflecting nonlinear population encoding of value (Fig. S5A). By contrast, in Monkey T, the trajectories were linearly separated and orderly, progressing orderly from low to high value offers (Fig. S5B).

6. Population decoding revealed differential integration of reward and delay.

We performed a moving window decoding analysis in which we decoded offer identity, reward size and delay for each trial based on spike count in each time bin (see Methods). For both animals, decoding accuracy was significantly above chance for all three variables (reward size, delay, and offer identity). Peak decoding accuracy reached nearly 80% for reward size and delay (chance is 33%), and over 60% for offer identity (chance is 11%) (Fig. 7, colored lines), indicating that all variables were robustly represented in dlPFC activity.

Next, we quantified the integration between reward and delay in dlPFC, by assessing whether the decoding accuracy for offer identity (i.e., the interaction between reward and delay) can be fully predicted by assuming the independence of coding for reward and delay. Specifically, if the dlPFC neural population encodes reward and delay independently, without integration, then the expected decoding accuracy for offer identity (i.e., the interaction between reward and delay), $\overline{P}(\text{offer identity})$, should equal the product of their individual decoding accuracies $P(\text{reward size})$ and $P(\text{delay})$:

$$\overline{P}(\text{offer identity}) = P(\text{reward size}) \cdot P(\text{delay})$$

If the observed decoding performance $P(\text{offer identity})$ exceeds the predicted level $\overline{P}(\text{offer identity})$, it would suggest that reward and delay are integrated and not represented independently. We compared the predicted decoding performance of offer identity $\overline{P}(\text{offer identity})$, with the actual decoding performance $P(\text{offer identity})$ for the two monkeys. For Monkey S, we found that $P(\text{offer identity})$ was significantly higher than $\overline{P}(\text{offer identity})$, from around 200ms after the offer onset (Fig. 7E). In contrast, no such difference was observed in Monkey T, suggesting that reward and delay were represented independently without integration in Monkey T (Fig. 7F). These results align with the respective behavioral strategies of the two monkeys.

Discussion

In this work, we trained two monkeys to perform a two-step temporal discounting task. They were required to make accept or reject decisions regarding delayed reward offers by releasing a lever when the purple ‘accept’ or yellow ‘reject’ go signal was presented. We showed that both of our monkeys adjusted their temporal discounting behavior based on the go signal. Specifically, when the first go signal was yellow (animals have to wait an additional delay after the yellow go signal to turn purple to accept the offer) compared to purple (animals can accept the offer immediately), both monkeys were less likely to accept the original delayed reward offer due to the additional wait time. Behavioral modeling revealed differences in strategy between monkeys. Monkey S recomputed the temporally discounted value by considering the additional delay associated with the go signal. In contrast, Monkey T treated the go signal as having additional additive value, without recomputing the temporally discounted value of the original offer. At the neural level, we found evidence of temporal discounting computation in dlPFC of both monkeys. The population encoding dimensions for reward and delay were inversely aligned in the dlPFC, in line with previous research (17, 18). Despite these similarities, however, we found that dlPFC neurons encoded reward and time in ways that reflected the behavioral differences between the animals. This suggests that the coding of reward and time in dlPFC depends on the strategy employed.

For monkey S, we observed that the population coding dimension for the go signal was positively aligned with the reward size coding dimension and negatively aligned with the delay coding dimension. This suggests that for monkey S, the go signal modified the discounted value, which is consistent with behavioral modeling results indicating that monkey S re-computed the temporally discounted value by considering the additional delay associated with the go signal. In contrast, for monkey T, the yellow/purple coding dimension was orthogonal to the value coding dimension, which aligned with the behavioral model in which monkey T assigned additive values to the go signal without re-computing the temporally discounted value.

Additional single-neuron and population-level analyses confirmed the neural differences between the animals in how dlPFC encoded reward and time. In the first analysis, we examined the linearity of value coding. In Monkey S, only 10% of the neurons tracked discounted value linearly; however, the population showed an early inverse correlation between reward magnitude and delay. Population trajectories were qualitatively entangled, suggesting that value integration was distributed across the population and involved complex nonlinear relationships with reward and delay. In contrast, in Monkey T, many more neurons (> 30%) showed linear value coding. Population trajectories were qualitatively unentangled and linear, with low-dimensional population trajectories clearly separated by value. In the second analysis, we assessed more directly whether reward and delay were represented independently. If reward and delay were encoded independently, we should be able to predict the identity of the offer identity (i.e., which of the nine delayed reward offers was presented to the animal) by decoding reward and delay separately. On the other hand, if reward and delay information were integrated non-linearly, then we would expect to be able to decode delayed offer identity more accurately than predicted by independence. We found statistical evidence that the decoding accuracy of offer identity exceeded the independence prediction for monkey S, but not for monkey T. These are consistent with the above analyses of coding linearity. Together, these results demonstrate that the coding of reward and time in dlPFC is strategy dependent.

The strategy-dependent coding that we observed is consistent with theories that dlPFC can operate in multiple computational modes (21), depending on the environmental demands and the cognitive strategy employed by the subject. This flexibility may be supported by mechanisms including 1) mixed selectivity (22), which enables neurons to process multiple task-related information (e.g., reward and delay), and 2) working memory gating (23), which selectively maintains task-relevant information during deliberation. In line with these mechanisms, dlPFC may also serve as a key node in the interaction between attention and motor planning (24, 25), providing the neural substrate that supports the distinct strategies. The neural differences observed between animals may arise from variations in prefrontal dopamine signaling and cognitive control (26). Our findings raise important questions about how prefrontal circuits may shift between different computational modes, and how these shifts may support adaptive or contribute to maladaptive behavior (27-29). Further research is needed to characterize the dynamical systems mechanisms underlying strategy shifts in temporal discounting (18).

Our findings also contribute to the recent efforts to identify the neural underpinnings of individual differences in behavior. For example, Pagan et al., (2025; 30) showed that rats can use different combinations of dynamic solutions to make context-dependent decisions, despite uniformly good task performance, resulting in heterogeneous neural and behavioral signatures. Similarly, Fascianelli et al., (2024; 31) found that, even when monkeys appear to perform a similar task, differences in the geometry of prefrontal neural representations reveal distinct strategies that can be linked to behavioral differences.

The behavioral differences in strategy observed in this study may be due to training history. Monkey S underwent more training with the task than Monkey T, and it is possible that with further training Monkey T's suboptimal strategy would have converged to the near-optimal strategy employed by Monkey S. Future longitudinal studies would help to clarify whether these strategies reflect innate individual traits or learned responses based on experience.

In summary, our results support strategy-dependent coding of reward and time in dlPFC during temporal discounting, and suggest that dlPFC activity reflects individual differences in temporal discounting strategies. These results also emphasize the importance of considering individual differences when studying the neural basis of behavior.

Methods

Subjects

Two male rhesus monkeys, both weighing 9.0 kg each were used. All experimental procedures followed the Institute of Laboratory Animal Research *Guide for the Care and Use of Laboratory Animals* and were approved by the National Institute of Mental Health Animal Care and Use Committee.

Surgical procedures

The surgical procedures were performed using sterile techniques and under general anaesthesia in a fully equipped operating room. The first surgery was performed to place a head fixation post. Monkey T received an eye coil, but eye movement was not monitored. A second surgery was performed to implant the Utah recording arrays. One Utah array was implanted in Monkey T, and two Utah arrays were implanted in Monkey S. MR scans were obtained 2-4 months after the Utah arrays were implanted to define the recording area (Fig. S1A).

Data collection

A non-commercial software package (REX, https://nei.nih.gov/intramural/software; 32) was used to control stimuli presentation, reward delivery and bar touches Eye position was monitored by using the Eye Position Measuring System (https://staff.aist.go.jp/k.matsuda/eye/indexe.html), but only during some sessions with Monkey S. Neural activity of single units was recorded using

Utah arrays (Blackrock Microsystems, Salt Lake City, USA; blackrockmicro.com). For Monkey T, electrical signals were captured, and single units were isolated online with a TDT (Tucker-Davis Technologies, TDT, Alachua, USA; tdt.com) system. For Monkey S, single units were isolated online with Grapevine system (Ripple Neuro, Salt Lake City, USA; rippleneuro.com) and then revised offline with Plexon offline sorting system. Data analyses were performed in MATLAB (http://www.mathworks.com/; version R2024b).

Task

The monkey sat in a primate chair with its head fixed 57 cm away from a video screen (47 x 30 cm; 1024 x 768 pixel; 120 Hz), that was placed in front of it. A touch-sensitive bar was mounted on the chair at the level of the monkeys' hands. A metal tube was positioned in front of the monkey's lips to allow the monkey to drink when a reward was delivered.

The monkey participated in an intertemporal discounting task (Fig. 1A). To start a trial, the monkey had to touch and hold a lever for 1000 ms until a red square (0.40º x 0.40º) appeared at the center of the screen. After a variable time of 250-500 ms one of 9 possible visual cues (9.2º x 7.3º) appeared at the center of the screen for one of five durations between 750 and 2000 ms (step size 313 ms). Each visual cue corresponded to a specific offer. These offers were made by combining one of three reward sizes (2, 4 or 6 drops of liquid reward) with one of three reward delivery delays (1, 5 or 10 seconds). The monkey was required to continue holding the lever and, if necessary, to maintain their gaze within the virtual window until the red square changed color to purple or yellow with equal probability (Fig. 1A). The purple or yellow go signals indicated that the monkeys could release the lever. Releasing the lever while the purple go signal was present indicated that the monkey accepted the offer. Releasing the bar when the yellow go signal was present indicated that the monkey refused the offer. If the monkey held the bar for the entire 1100 ms purple or yellow period, the color of the go signal would change, allowing the monkey to accept or refuse later in the trial.

For example, if the monkey did not release the lever during the initial purple go signal period, it could hold the bar until the yellow go signal replaced the purple one. The same thing happened if the monkey did not release the lever during the initial yellow go signal period. It could hold the bar until the purple go signal replaced the yellow one and then release the bar. With this design, we prevented the monkey from anticipating whether the acceptance or refusal period would be the first interval, so it could not preprogram the motor response in the cue period.

If the offer was accepted, the purple go signal turned blue (the feedback signal), and the reward drops were delivered after the delay. If the offer was refused, the cue and fixation target disappeared, and a new trial would begin immediately with a new offer.

The eye of Monkey S was monitored for 12 out of 16 sessions. A virtual window 20 x 20 degrees was built around the cue. The monkey was required to maintain his gaze within the virtual window until the go signal appeared.

An error was registered if the monkey (a) released the lever during the cue period, (b) did not maintain gaze in the virtual window, or (c) failed to release the lever during any of the go periods. In this case, an error feedback image (a white rectangle with a red cross measuring 28.7^{o} x 17.8^{o}) appeared on the screen for 1000 ms. Following an error, the next trial was a correction trial, during which the offer was repeated.

Behavior

For the behavioral analyses, we used data from all the neural recording days (7 for monkey T and 16 for monkey S).

The probability of accepting an offer was calculated for each of the 9 offers independently. The behavior of the two monkeys was modeled separately using reinforcement learning models, in which the probability of accepting an offer was modelled through a *SoftMax* function as

$$p = \frac{1}{1 + e^{\beta \cdot (\theta - V(drop, delay))}}$$

Here, θ is the decision threshold and β is the temperature term.

$$V(drop, delay) = \frac{drop}{1 + k \cdot delay}$$

k is the hyperbolic discounting factor. The free parameters θ, β, k were fitted via maximum likelihood estimation.

To examine how the timing of go signals shapes value computation, and whether monkeys show different strategies, we compared three behavioral models.

1. Model 1

Model 1 computes a temporally discounted value, that is then compared to an accept threshold θ.

$$DV = \frac{R}{1 + k \cdot D}$$

Here, DV is the discounted value, D is the delay, R is the reward.

Animals will be more likely to accept when DV is above the acceptance threshold θ, and more likely to reject otherwise. The probability of accepting an offer is modelled using the softmax rule with decision noise σ:

$$p(accept) = \frac{1}{1 + e^{\frac{-V - \theta}{\sigma}}}$$

In this situation, animals do not pay attention to the go signal and choose solely based on the offer.

2. Model 2

In the model 2, animals simply have a bias for the color of the go signal, purple or yellow.

$$p(acceptpurple) = \frac{1}{1 + e^{\frac{-U-\theta+\beta}{\sigma}}}$$

whereas

$$p(acceptyellow) = \frac{1}{1 + e^{\frac{-U-\theta-\beta}{\sigma}}}$$

3. Model 3

In the model 3, animals adjust their discounted value estimate as well as their acceptance threshold based on the go signal. If the go signal is yellow, they have to wait for an additional delay Δt before they can accept the offer, as a result, effectively the value of the offer decreases to

$$V^{yellow} = \frac{R}{1 + k \cdot (D + \Delta t)}$$

Since threshold effectively reflects the average expected discounted reward during this task, the total value of choosing to accept a yellow offer becomes

$$U^{yellow} = V^{yellow} + VF^{yellow}$$

Where VF^{yellow} is the expected future reward, and is θ discounted by the additional waiting time Δt plus the delay:

$$VF^{yellow} = \frac{\theta}{1 + k \cdot (D + \Delta t)}$$

The acceptance threshold would remain at $\theta^{yellow} = \theta$

However, if the go cue is purple, we compute

$$U^{purple} = V^{purple} + VF^{purple}$$

Since they can accept right away, $V^{Purple} = V$ and $VF^{purple} = \frac{\theta}{1+k \cdot D}$,

Rejecting is actually more expensive after purple, as animals have to wait for an additional delay to reject. The acceptance threshold here is the expected discounted value of future rewards after a rejection. This would be:

$$\theta^{purple} = \frac{\theta}{1 + k \cdot \Delta t}$$

In either case,

$$p(accept) = \frac{1}{1 + e^{\frac{-U-\theta}{\sigma}}}$$

ANOVA-sliding window Analysis

A sliding window analysis was performed using the neural activity from all 170 recorded cells, expressed as total spike count. Starting 500 ms before and ending 1000 ms after visual cue onset, 100 ms bins were created with a 10 ms interleave period. The spike count within each bin was used in the two-way ANOVA sliding window analysis, with drops and delay as the independent variables. An additional effect was included for the choice factor to test whether choice explains additional variance in firing rate, independently of drop and delay.

For each bin, we estimated the proportion of cells that were significantly modulated by each factor (two-way ANOVA, $p < 0.05$).

Additionally, we extracted the variance explained by each factor, and by the interaction between drops and delay, expressed as omega squared (33):

$$\omega^2 = (SS_{effect} – (df_{effect})(MS_{error}))/ MS_{error} + SS_{total}$$

Correlation across time

For this analysis, we only included cells that responded to at least one of the task-relevant variables (delay, drop or choice). This left 121 cells.

To extract the population encoding dimension for drop and delay, we performed a sliding window ANOVA on each cell and extracted the regression coefficients for drop and delay at each time bin. The dimension formed by the beta coefficients across cells at each time bin, is the time-dependent encoding dimension of drop or delay.

The population activity that encodes delay is quantified as the activity projected onto the delay dimension. The population activity of drops is calculated similarly. For each factor separately, the beta coefficients for each time bin were correlated with each other, and the Pearson correlation was calculated.

A significance threshold of 0.05 (uncorrected) was used.

Decoding analysis

A support vector machine was used as the classifier. Training and testing comprised of non-overlapping trials (75% of trials for training, and 25% for testing). This procedure was repeated 10 times, and the average decoding accuracy was recorded. Decoding of drops, delays and offer identity were carried out separately. For each factor, decoding was performed using a sliding window (step size: 20 ms, width: 100 ms).

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Additional information

Data availability

Data will be available once the manuscript has been accepted.

Conflicts of interest

All authors should disclose any competing interests/conflicts of interest in accordance with journal's policy.

Author contributions

DW, BR and RF conceived and designed the task. RF and DW collected the data. SW, RF, ME, and BR analyzed the data. MMc was involved in the initial analysis of the data. RF, DW and ME performed the surgeries and MRIs. RF, SW, ME and BR wrote the paper.

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Figures

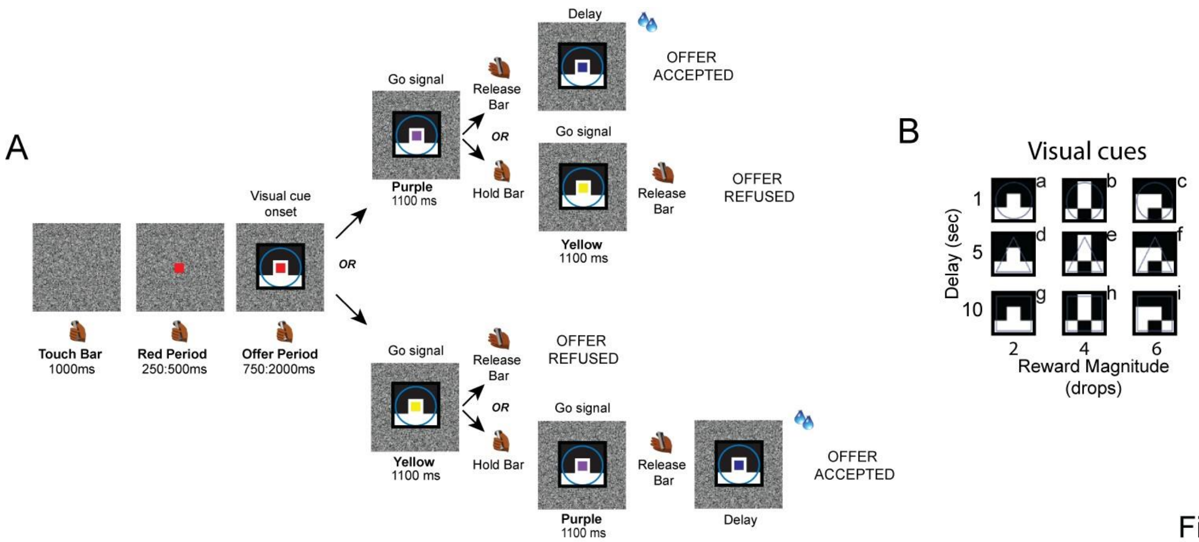

Fig. 1

Figure 1. Task description. (A) Each trial begins when the monkey touches and holds a bar. A red square then appears in the center of the screen, followed by a visual cue presented behind the red square. This cue indicates the offer, which is defined by a combination of reward magnitude (2, 4, or 6 drops) and delay (1, 5, or 10 s). Next, the red square changes color to either purple or yellow, serving as a go-signal. If the square turns purple, the monkey can release the bar to accept the offer. The reward is then delivered after the indicated delay. Alternatively, the monkey can continue holding the bar until the purple spot turns yellow to refuse the offer. If the square turns yellow, the monkey can release the bar to refuse the offer and start a new trial immediately. Alternatively, the monkey can continue holding the bar until the yellow turns purple to accept the offer.

(B) The set of nine visual cues used in the task are labeled from “a” to “i”.

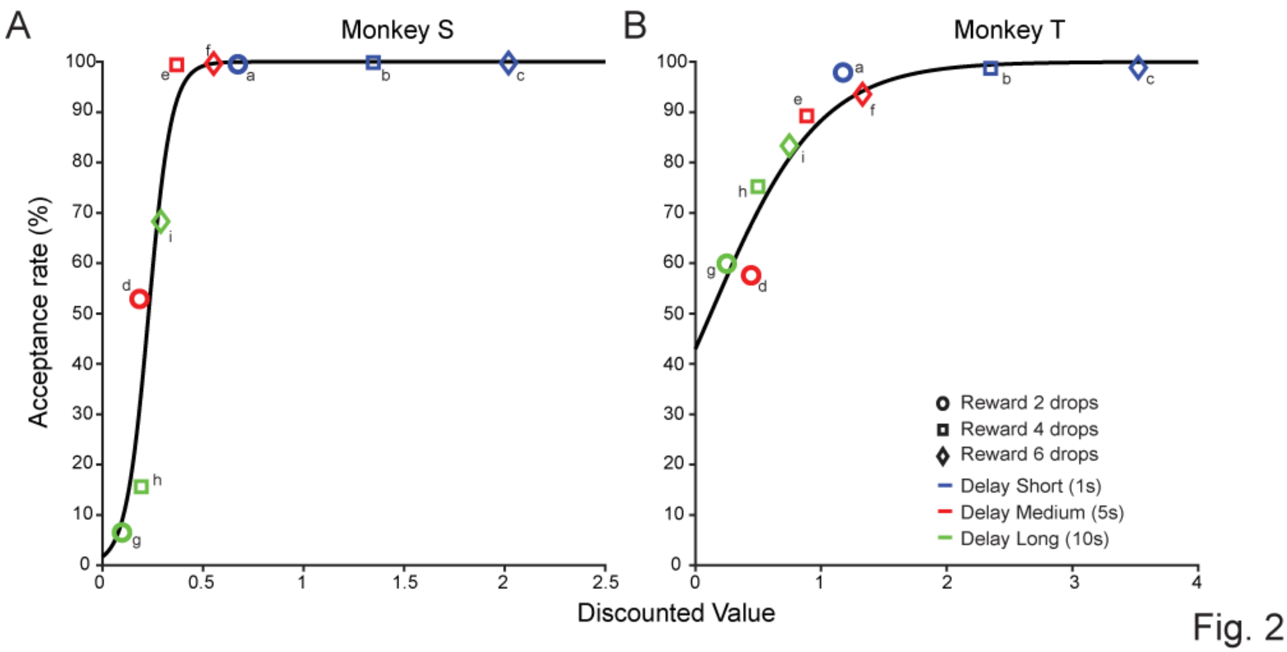

Figure 2. Behavioral performance. Acceptance rate as a function of the discounted value inferred by the model (data: colored points; model: black line) in Monkey S **(A)** and Monkey T **(B)**. The choices were fitted with hyperbolic discounting and logistic functions to estimate the discounted value and acceptance probability, respectively. Letters from “a” to “i” indicate the nine possible offers. In this analysis the total delay was the cue-indicated delay (1 s, 5 s or 10 s) and the reward magnitude was the cue-indicated number of drops (2, 4 or 6).

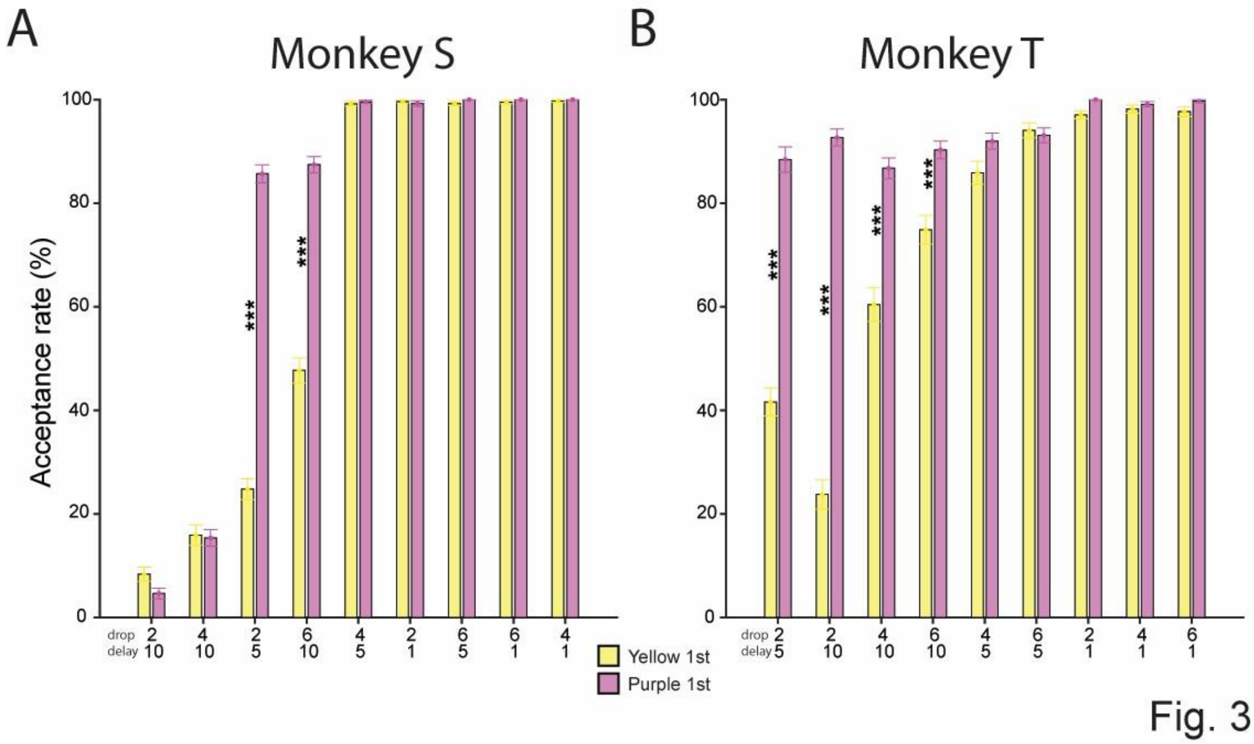

Figure 3. Acceptance rate during the first go signals. Acceptance rate for each offer when the first go signal was yellow or purple, shown for Monkey S (**A**) and Monkey T (**B**). The offers are sorted by their empirical acceptance rate across the nine cues.

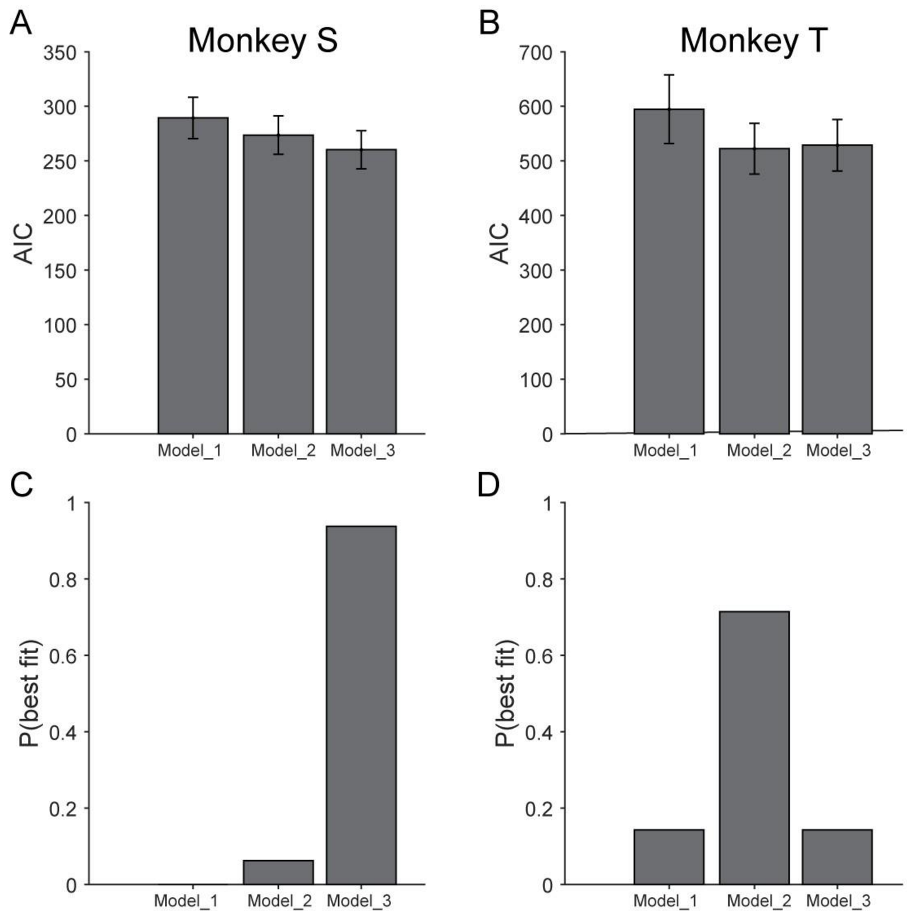

Fig. 4

Figure 4. Behavioral Models. Model comparison of AIC values (A and B) and probability of best fit (C and D) for the three behavioral models in Monkey S (A and C) and Monkey T (B and D).

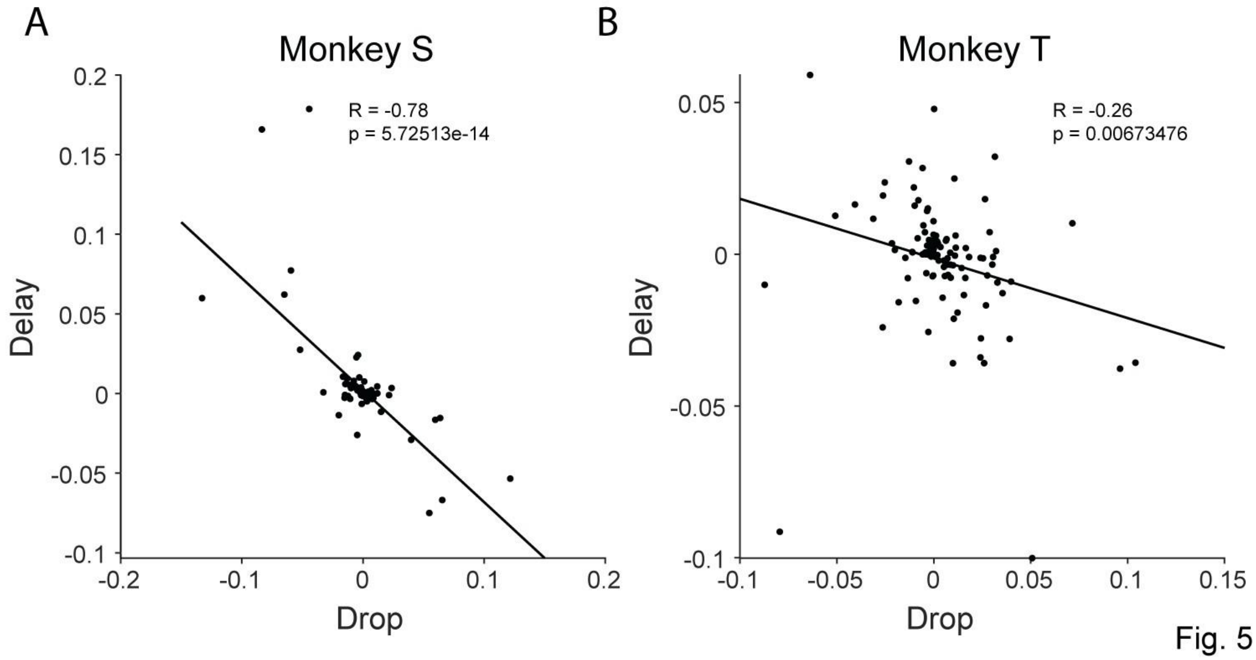

Figure 5. Correlation of drops and delay population encoding dimensions. The population encoding dimensions of drops and delays are negatively correlated in Monkey S (**A**) and Monkey T (**B**). The population encoding dimensions were estimated using a time window from 750 ms to 1000 ms after the visual cue onset.

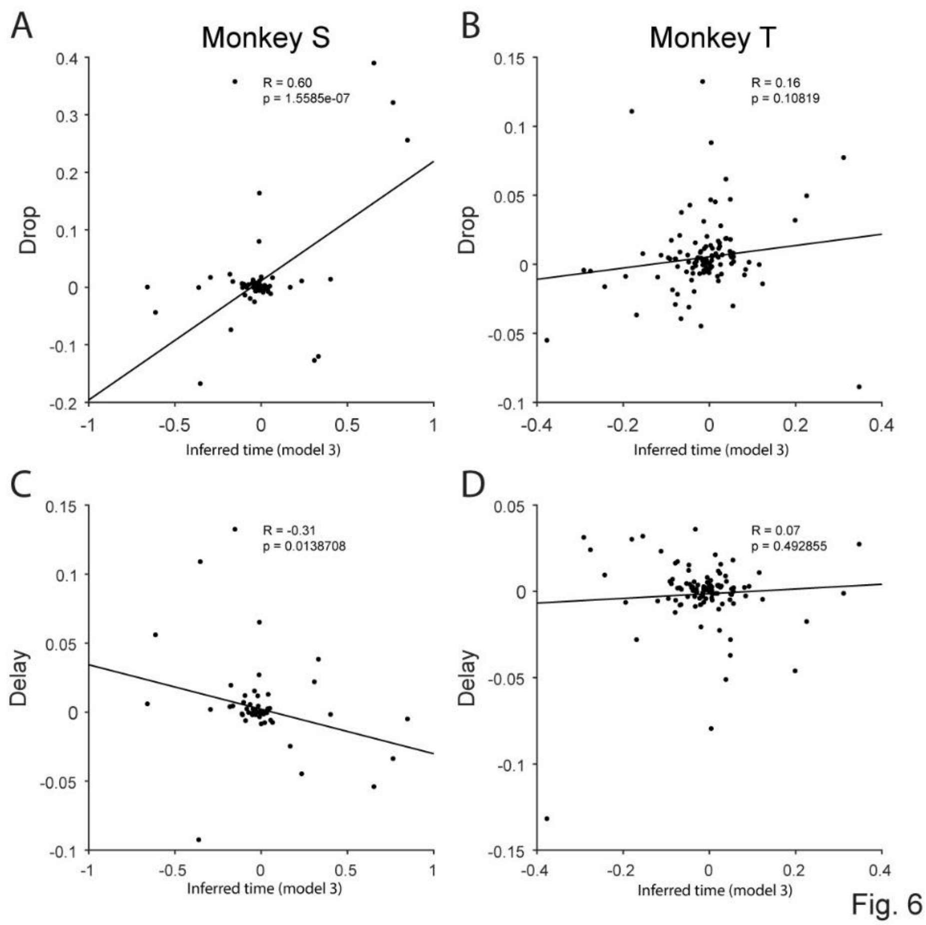

Figure 6. Linear correlation between drops and model 3 inferred time (see methods) neural encoding. Top panels: correlation between drops and inferred time from Model 3 in Monkey S **(A)** and Monkey T **(B)**. **Bottom panels:** correlation between delay and inferred time from Model 3 in Monkey S **(C)** and Monkey T **(D)**.

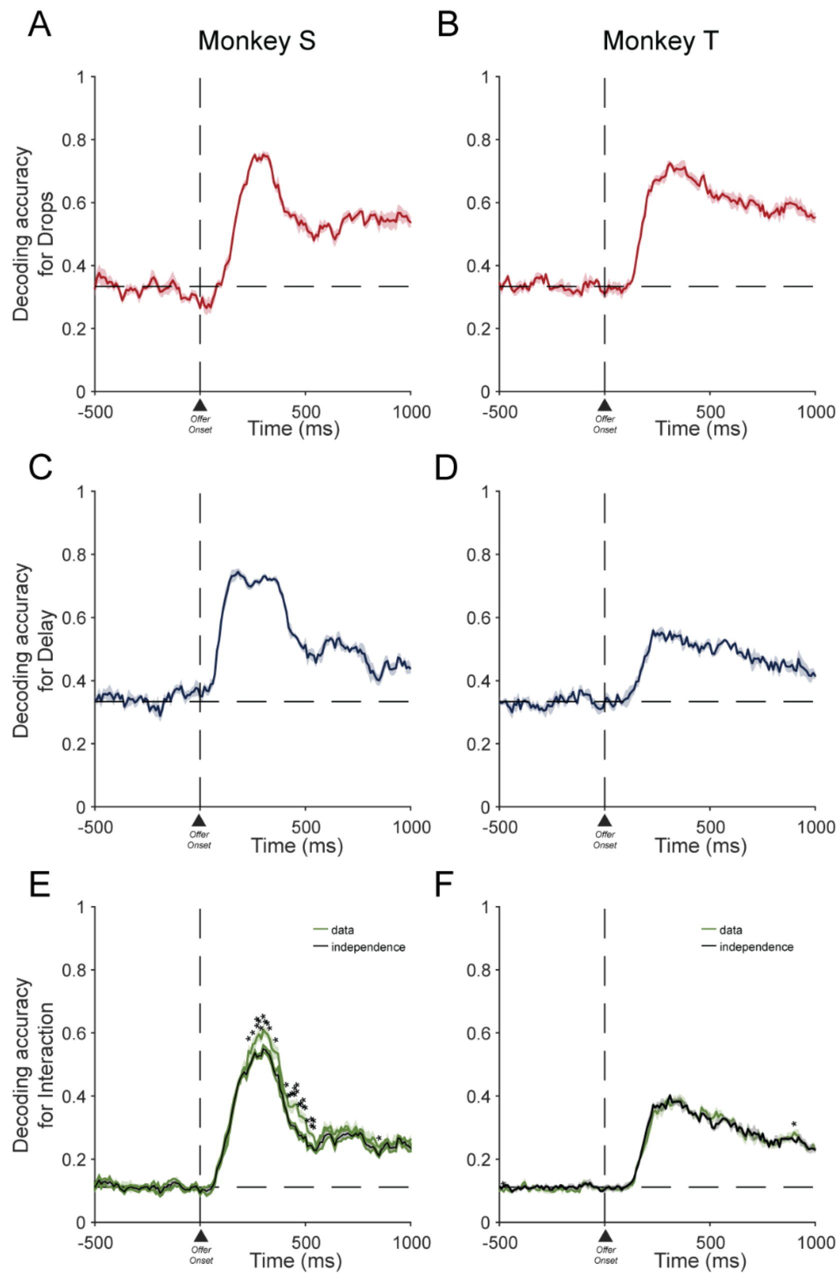

Fig. 7

Figure 7. Decoding Analysis. Decoding accuracy was calculated for each task factor: drops (top panels), delay (middle panels), and interaction (bottom panels). This was done for both Monkey S (panels A, C and E) and Monkey T (panes B, D and F), from 500 ms before to 1000 ms after the offer onset.

Supplementary Information

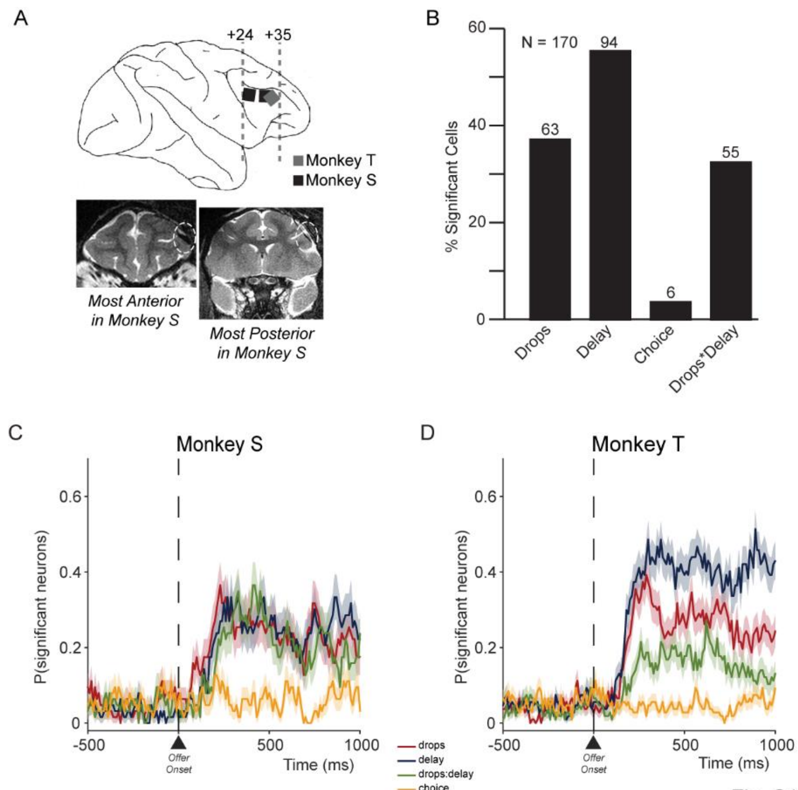

Figure S1. Recording sites and neural modulation by task factors. (A) Schematic representation of the Utah array positions in the dlPFC, reconstructed from MRI scans of Monkeys S and T. Bottom panel: the most anterior and posterior MRI slices of Monkey S, with the Utah arrays highlighted by yellow dashed circles. **(B)** Percentage of neurons that were significantly modulated by reward drops, delay, their interaction, and upcoming choice. **(C–D)** The time course of the percentage of neurons modulated significantly by each factor, from 500 ms before to 1000 ms after cue onset, in monkeys S and T, respectively.

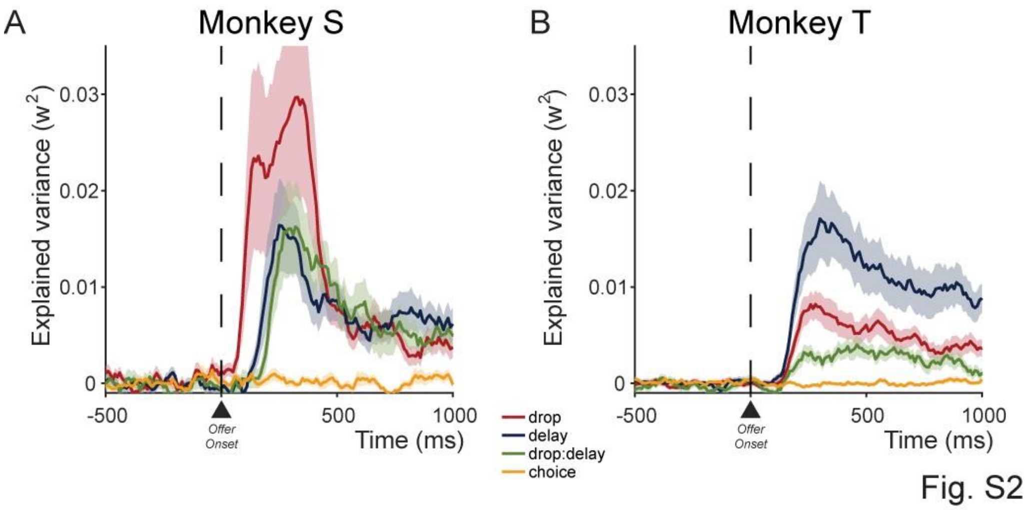

Figure S2. Explained variance. The variance explained is calculated for each task factor over time, from 250 ms before to 1000 ms after the cue onset in Monkey S (**A**) and Monkey T (**B**).

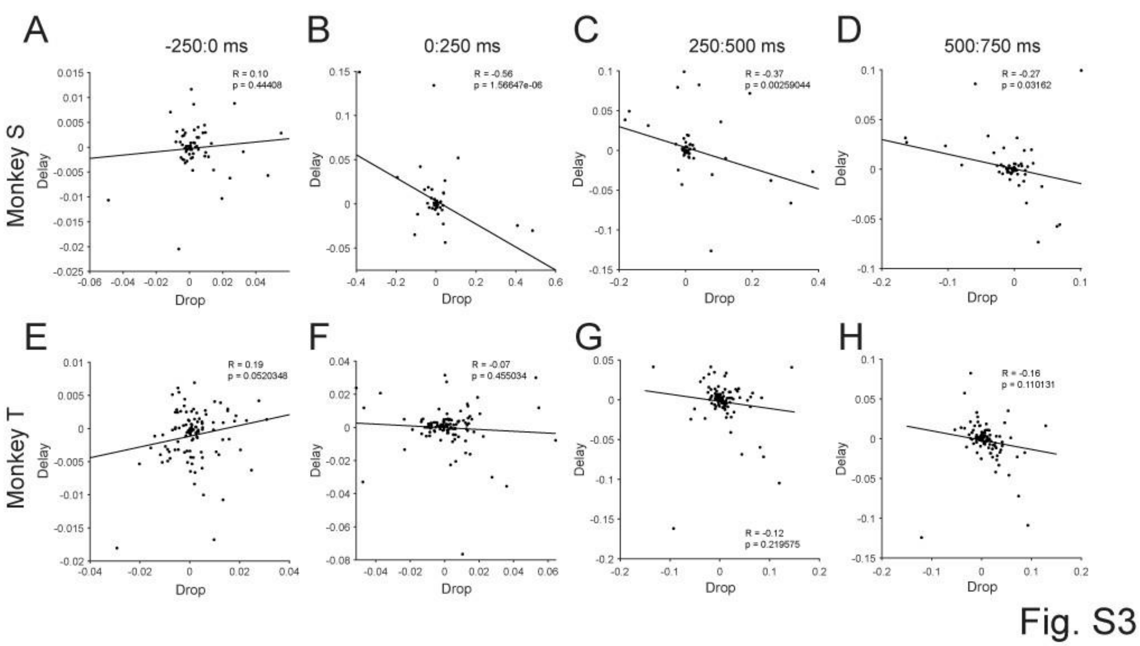

Figure S3. Linear correlations between drops and delay neural encoding dimensions. The population encoding dimensions for drops and delays were calculated in 250 ms time bins, before and after the visual cue onset (time 0) for Monkey S (A–D) and Monkey T (E–H).

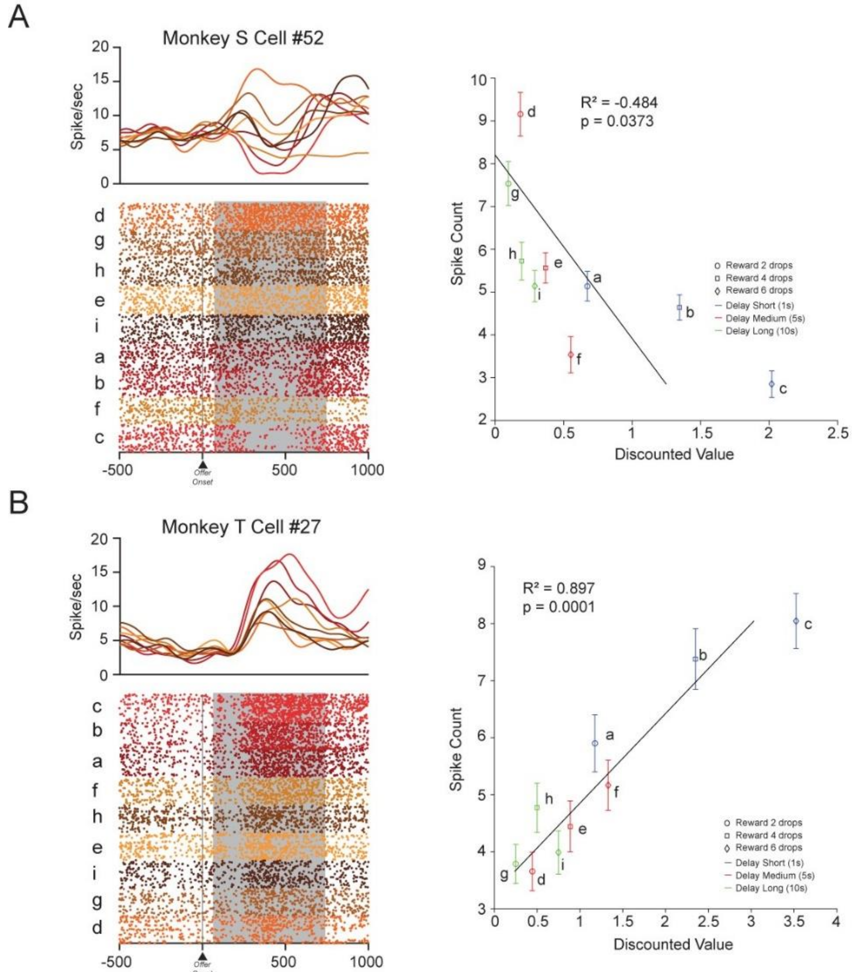

Fig. S4

Figure S4. Raster plots and correlation between mean spike count and discounted value. **Left panels:** example neurons from Monkey S (**A**) and Monkey T (**B**). Spike density functions are shown above each raster. The neural activity is aligned to the cue onset and divided into nine offers, which are ordered from the highest to the lowest mean firing rate. The mean firing rate is calculated from 75 to 750 ms after the cue onset (shown by the grey rectangle). **Right panels:** the linear correlation between the mean spike count for each offer and the behaviorally calculated discounted value, corresponding to the neurons shown in the left panels.

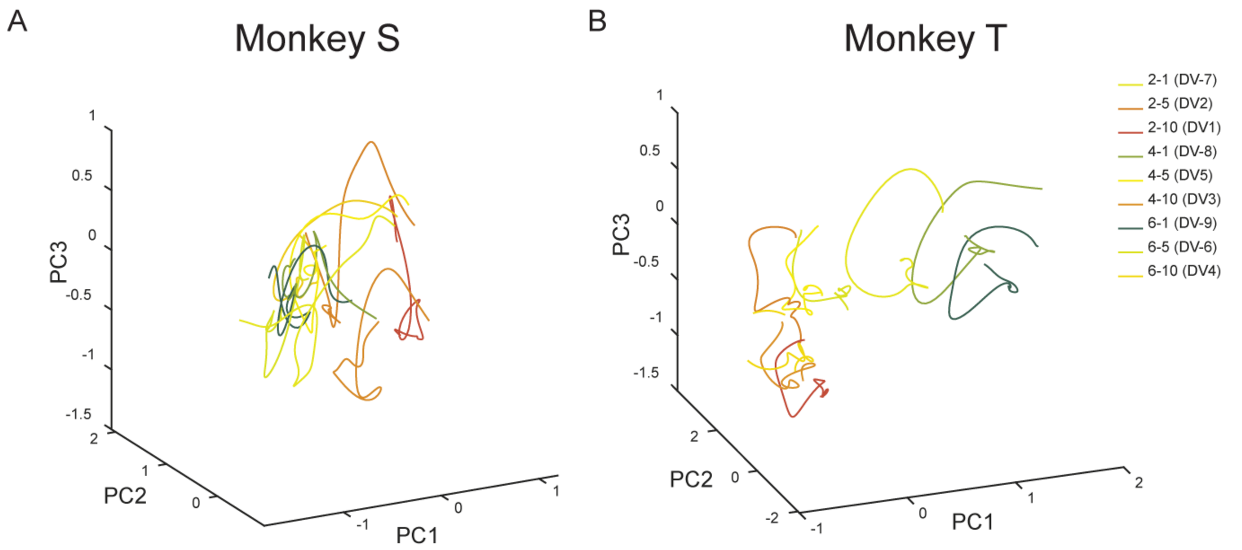

Fig. S5

Figure S5. Geometry of neural coding. Population trajectories for each offer in Monkey S (A, C) and Monkey T (B, D). The trajectories are plotted in the 3D spaces derived from a principal component analysis performed over a period of 500 ms before and after the first go signal onset.

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Title: Strategy-dependent coding of reward and delay during temporal discounting in macaque dorsolateral prefrontal cortex

Authors: Rossella Falcone[1,3,†,*], Siyu Wang[1,†], David B. Weintraub[2], Mariko McDougall[1], Mark A. G. Eldridge[1,4], and Barry J. Richmond[1,5].

Author affiliations:

[1]Laboratory of Neuropsychology, NIMH, Bethesda, MD, USA

[2]Surgical Neurology Branch, NINDS, Bethesda, MD, USA

[3]Leo M. Davidoff Department of Neurological Surgery, Albert Einstein College of Medicine Montefiore Medical Center Bronx, NY USA.

[4]Newcastle University, Newcastle upon Tyne, NE1 7RU.

[5]Senior Investigator, Emeritus.

†Equal contribution.

***Corresponding author:**

Rossella Falcone, PhD

Assistant Professor

Albert Einstein College of Medicine, Bronx, NY, 10461

e-mail: rossella.falcone@einsteinmed.edu

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